(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

FIFTH SEMESTER [B.TECH] JANUARY 2024

Paper Code: ECC-305 **Subject: Microelectronics**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Each question carry equal marks. Assume suitable missing data if any.

Q1 (a) Differentiate between enhancement and depletion type MOSFET. Which type is a preferred choice and why? Support your answer with relevant diagrams. **(2.5)**

(b) What is Moore's Law? With downscaling of device dimensions by a factor of 10, what would you expect for drain current will be ---- and --- times of the original value if constant field scaling and constant voltage scaling are used. **(2.5)**

(c) Draw the static CMOS logic circuit for the following expression: **(4)**
(i) Y= (AB'CD)' (ii) Y= [D(A+BC)]'

(d) Write short notes on: **(1x6=6)**
(i) Channel Length Modulation
(ii) Ferroelectric RAM
(iii) Threshold Voltage
(iv) Twin Tub Process
(v) Impact of increasing threshold voltage of NMOS transistor on CMOS VTC in terms of noise margins (VIL, VIH).
(vi) CMOS vs BJT

UNIT-I

Q2 (a) An NMOS transistor, operating in the linear region with VDS = 50 mV, is found to conduct 25 μA for VGS = 1 V and 50 μA for VGS = 1.5 V. What is the apparent value of threshold voltage VT? If kn' = 50 μA/V2, what is the device W/L ratio? What current would you expect to flow with VGS = 2 V and VDS = 0.1 V? If the device is operated at VGS = 2 V, at what value of VDS will the drain end of the MOSFET channel just reach pinch-off? **(7.5)**

(b) Explain in detail a typical N-well CMOS fabrication process. Support it with suitable diagrams. **(7.5)**

Q3 (a) With the help of energy band diagram, explain the operation of MOSFET under external bias. **(7.5)**

(b) Derive the expression for IDS for NMOS transistor in non-saturation region and draw IDS vs VDS characteristics for enhancement mode MOS transistor. **(7.5)**

UNIT-II

Q4 (a) Design a CMOS inverter with switching threshold voltage of 2 /3 *VDD*. What is the resulting ratio of *WP /WN* ? (Assume LP = LN, μN = 2.5μP, |VTP|=VTN = 0.2 VDD). **(2.5)**

(b) Design the following Boolean functions using multiple output domino logic: **(5)**
C1 = G1 + P1C0, C2 = G2 + P2G1+ P2P1C0, where Pi = Ai Oi +AiOi , Gi = Ai Oi (i = 1,2,3)

(c) What is a pass transistor? Explain the different regions of operation of CMOS inverter. Support with a diagram. **(5)**

(d) Explain CMOS based NOR SR latch circuit and its working. **(2.5)**

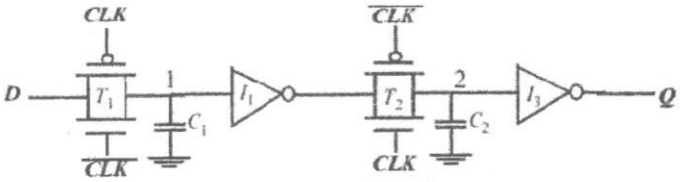
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Q5 (a) Define Latch-up. Is it desirable? Give the ways to reduce the same in CMOS circuits. **(4)**

(b) Discuss dynamic CMOS logic. Consider a three input NOR gate. **(5)**

(c) An implementation of a D Flip-flop is show in Fig. Answer the following: **(6)**
(i) Is this a static or dynamic flip-flop?
(ii) Is this positive edge triggered or negative edge triggered?
(iii) Calculate tsetup, tCtoQ and thold for this flop in terms of the transmission gate and inverter delays.



UNIT-III

Q6 (a) Explain the difference between cascade current mirror and Wilson current mirror with diagrams. **(7.5)**

(b) Explain the compensation of op-amps. **(7.5)**

Q7 (a) Develop CMOS Differential amplifier and obtain the differential Transconductance of the same. **(7.5)**

(b) Derive the expression for power-supply rejection ratio of Two-stage op-amps. **(2.5)**

(c) What is a current mirror circuit? Explain with the help of a circuit. **(5)**

UNIT-IV

Q8 (a) Explain read/write operation of SRAM memory cell. How 1 bit cell is used in bigger memory systems. **(7.5)**

(b) Distinguish between SRAM and DRAM. **(7.5)**

Q9 (a) Implement 2:1 MUX using CMOS transmission gate. **(5)**

(b) Write short notes on:
(i) Analog to Digital Converters (ADCs) **(5)**
(ii) Digital to Analog Converters (DACs) **(5)**

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